

Full Stack Dart

DroidCon London - October 2021





Flutter



A brief introduction



Engineer, The @ Company,
building a platform that puts people in
control of their data.

Co-host, Tech Debt Burndown Podcast

Cloud Editor, InfoQ

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Links to socials etc. at chris.swanz.net



Agenda

- Why Dart?
- Language features
- JIT vs AOT
- Dart on Docker
- Functions Framework for Dart
- Profiling and performance management
- Other places you can learn more
- Call to action - try out the Functions Framework Examples



Why Dart?



Dart

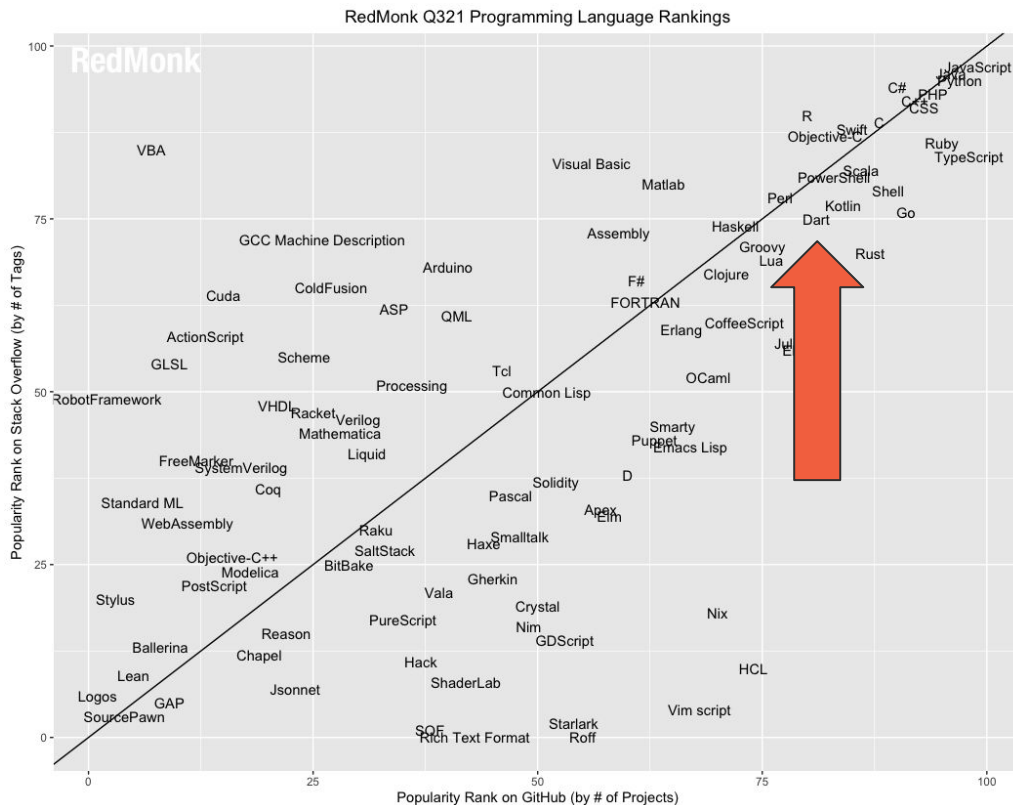
“Dart is a client-optimized language
for fast apps on any platform”

dart.dev



Flutter pulled Dart into the Redmonk Top 20

- 1 JavaScript
- 2 Python
- 2 Java
- 4 PHP
- 5 CSS
- 5 C++
- 5 C#
- 8 TypeScript
- 9 Ruby
- 10 C
- 11 Swift
- 12 R
- 13 Objective-C
- 14 Shell
- 14 Scala
- 16 Go
- 17 PowerShell
- 18 Kotlin
- 19 Rust
- 20 Dart



Language features

'Hello World!'

```
void main() {  
    print('Hello, World!');  
}
```



Garbage antipattern (just like Java or C#)

```
// Create an object of arguments to pass
var args = ['parent-element', 'child-element'];
// Create a result object and call function
var value = getValue(args);
// Null test value of result object
if (value != null) {
    // Return result object
    return value;
}
```



Async/await (like JavaScript)

```
Future<String> createOrderMessage() async {  
    var order = await fetchUserOrder();  
    return 'Your order is: $order';  
}
```

```
Future<String> fetchUserOrder() =>  
    // Imagine that this function is  
    // more complex and slow.  
    Future.delayed(  
        const Duration(seconds: 2),  
        () => 'Large Latte',  
    );
```

```
Future<void> main() async {  
    print('Fetching user order...');  
    print(await createOrderMessage());  
}
```



Concurrency - Isolates

“independent workers that are similar to threads but don’t share memory, communicating only through messages”



Sound null safety (since Dart 2.12, Mar 2021)

```
// In null-safe Dart, none of these can ever be null.  
var i = 42; // Inferred to be an int.  
String name = getFileName();  
final b = Foo();  
  
// To indicate that a variable might have the value null  
// just add ? to its type declaration  
int? aNullableInt = null;
```



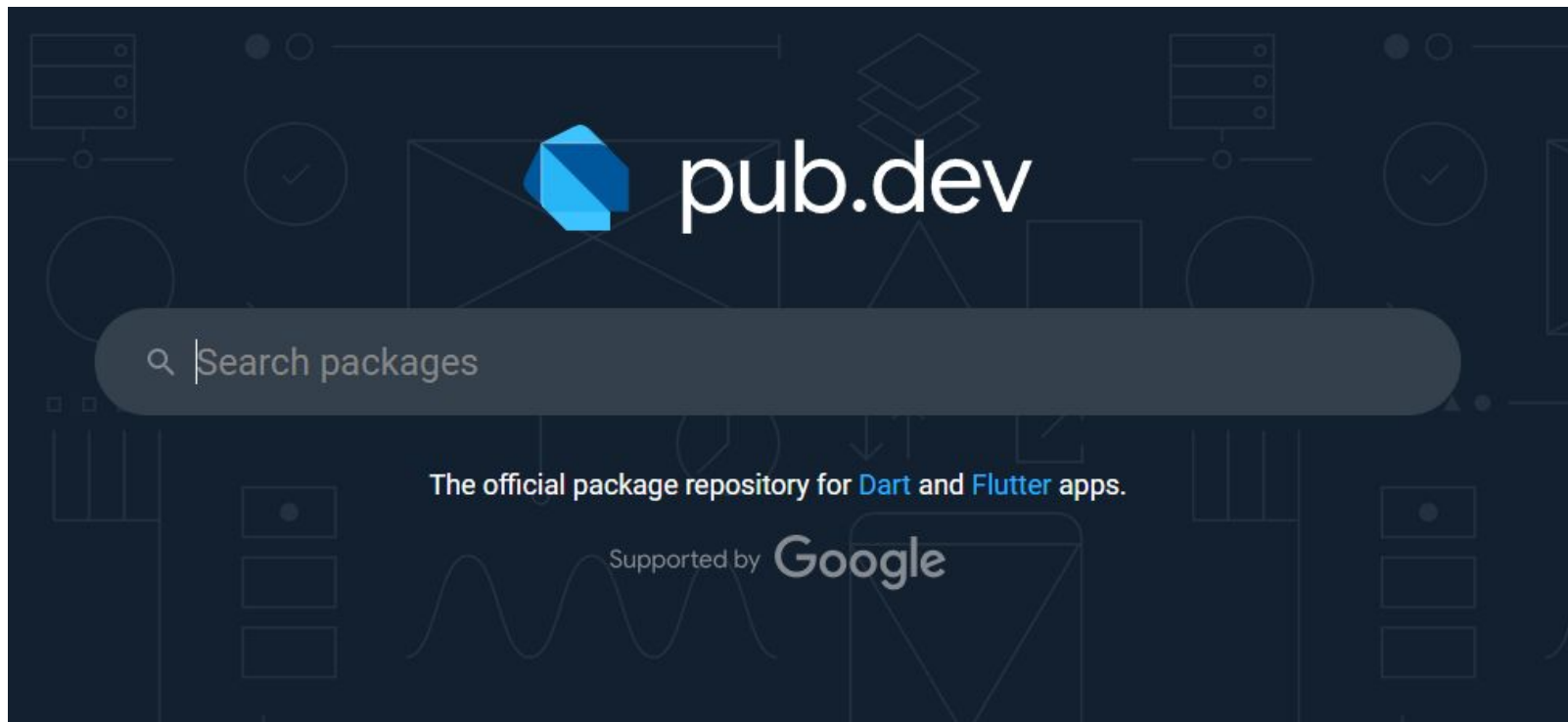
Licensing

BSD

license BSD3



Package Manager - pub.dev



JIT vs AOT

Dartshowplatform - A more useful 'Hello World!'

```
import 'dart:io' show Platform, stdout;  
  
void main() {  
    print(Platform.version);  
}
```

```
2.14.4 (stable) (Wed Oct 13 11:11:32  
2021 +0200) on "linux_arm64"
```



JIT - Just `dart run` it in the virtual machine

```
$ time dart run showplatform.dart
```

```
2.14.4 (stable) (Wed Oct 13 11:11:32  
2021 +0200) on "linux_arm64"
```

```
real    0m6.280s  
user    0m5.050s  
sys     0m0.915s
```



AOT - Compile it first then run the binary

```
$ dart compile exe showplatform.dart  
$ time ./showplatform.exe
```

```
2.14.4 (stable) (Wed Oct 13 11:11:32  
2021 +0200) on "linux_arm64"
```

```
real    0m0.028s  
user    0m0.019s  
sys     0m0.009s
```



Trade off - compilation is slow

```
$ time dart compile exe \  
showplatform.dart
```

```
Info: Compiling with sound null safety  
Generated: showplatform.exe
```

```
real    0m17.434s  
user    0m20.912s  
sys     0m2.958s
```



Dart on Docker

A typical Dockerfile for AOT compiled Dart binary

```
FROM dart AS build
WORKDIR /app
COPY ./mydemoapp.dart .
RUN dart pub get
RUN dart compile exe /app/mydemoapp.dart \
    -o /app/mydemoapp

FROM scratch
COPY --from=build /runtime/ /
COPY --from=build /app/mydemoapp /app/mydemoapp
ENTRYPOINT ["/app/mydemoapp"]
```



Image sizes for a trivial application



atsigncompany/dartshowplatform ☆

By [atsigncompany](#) • Updated 6 days ago

Multi arch Dart test image that prints out version and platform info

Container

TAG

[latest](#)

`docker pull atsigncompany/dartshowpl...`



Last pushed 6 days ago by [atcpswan](#)

DIGEST

[8f4d6b9d1d68](#)

[100829c90d10](#)

[1dac2840cb6e](#)

OS/ARCH

linux/amd64

linux/arm/v7

linux/arm64

COMPRESSED SIZE ⓘ

4.45 MB

3.26 MB

3.68 MB



Image sizes for a non trivial application



atsigncompany/secondary ☆

By [atsigncompany](#) • Updated an hour ago

@protocol secondary server

Container

TAG

[dess](#)

`docker pull atsigncompany/secondary:...`



Last pushed 2 days ago by [atcpswan](#)

DIGEST

OS/ARCH

COMPRESSED SIZE ⓘ

[197123ec5e7a](#)

linux/amd64

5.97 MB

[1a75e98e541f](#)

linux/arm/v7

5 MB

[6b20a465d6cd](#)

linux/arm64

5.21 MB



Functions Framework for Dart

GoogleCloudPlatform/functions-framework-dart

The Functions Framework lets you write lightweight functions that run in many different environments, including:

- Your local development machine
- Google Cloud Run - see cloud run quickstart
- Google App Engine
- Knative-based environments

Google Cloud Functions does not currently provide an officially supported Dart language runtime, but we're working to make running on Google Cloud Run as seamless and symmetric an experience as possible for your Dart Functions Framework projects.



examples/hello/lib/functions.dart

```
import 'package:functions_framework/functions_framework.dart';  
import 'package:shelf/shelf.dart';  
  
@CloudFunction()  
Response function(Request request) => Response.ok('Hello, World!');
```

To:

```
$ curl https://"YOUR_APP_URL"  
Hello, World!
```

All without needing to worry about writing an HTTP server or request handling logic.



Profiling and performance management

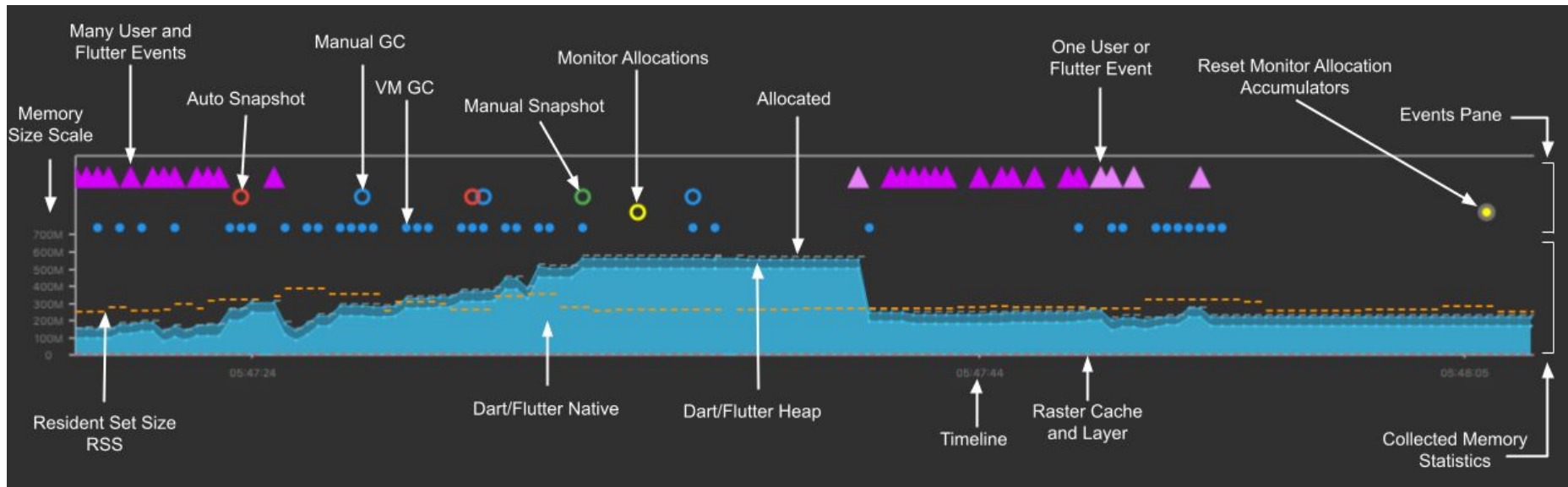
Dart DevTools

	Flutter mobile or desktop 	Flutter web 	Other web	Command-line
Debugger 	✓	✓	✓	✓
Logging view 	✓	✓	✓	✓
App size tool 	✓			✓
CPU profiler 	✓			✓
Memory view 	✓			✓
Network view 	✓			✓
Performance view 	✓			✓
Flutter inspector 	✓	✓		

<https://dart.dev/tools/dart-devtools>



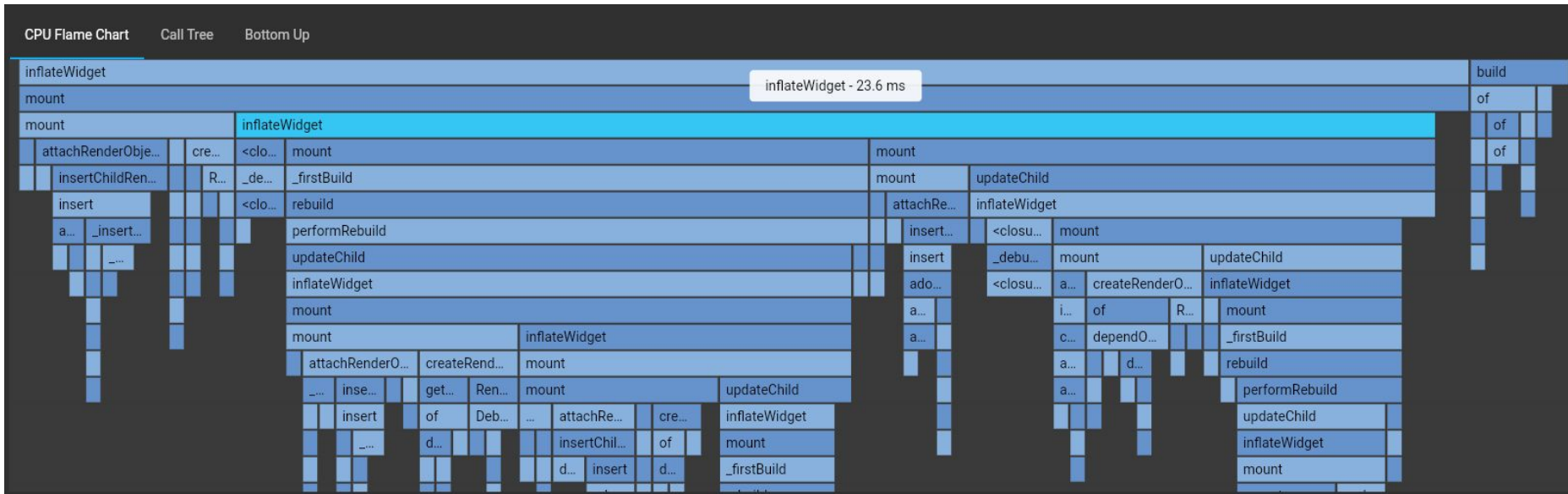
Memory view



<https://flutter.dev/docs/development/tools/devtools/memory>



Flame chart



<https://flutter.dev/docs/development/tools/devtools/performance>



But... DevTools need to connect to a VM, so JIT only

[@mraleph](#) where would you start with trying to track down a potential memory leak in an AOT Dart app (running in a container)?

I'm presently planning to run an instance of it in a JIT VM so I can use the memory profiler, but if there's a way to get metrics from an AOT binary?

6:20 pm · 12 Aug 2021 · Twitter Web App



Vyacheslav Egorov @mraleph · 12 Aug



Replying to [@cpswan](#)

yeah, this is a bit complicated - if it was a Flutter application you could use so called profile mode (AOT with profiling tools enable) to collect a heap profile. but dart compile exe always produces release mode snapshots.



1



1



Vyacheslav Egorov @mraleph · 12 Aug

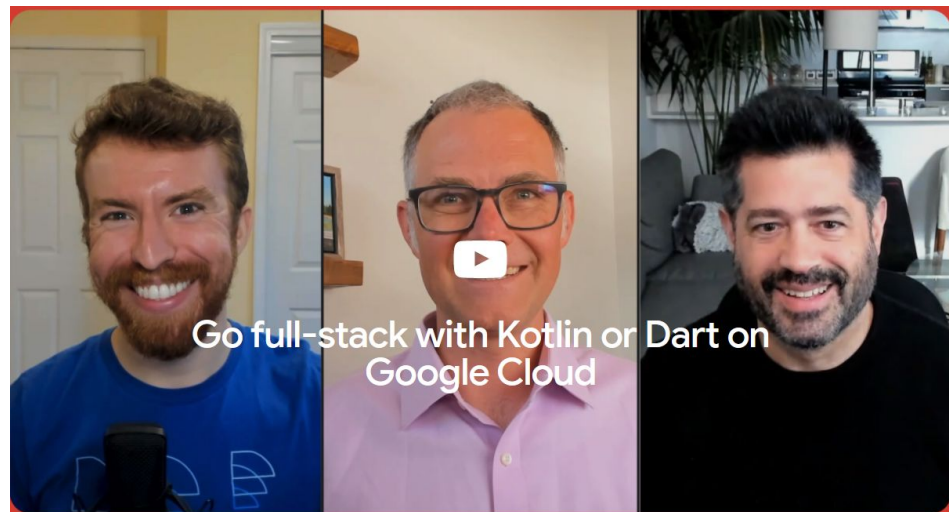
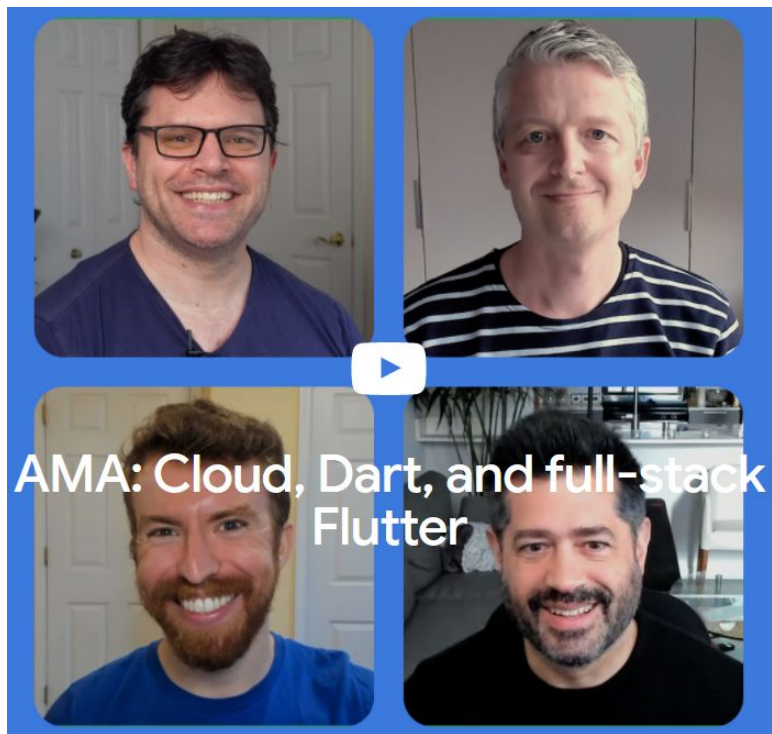


if you feel adventurous you could build your own SDK and change dart compile to produce vmervice enabled binaries.



Other places where you can learn more

<https://events.google.com/io/session/083b441e-4076-4443-9750-6529b34257b7?lng=en>



<https://events.google.com/io/session/6d5795f2-ec1c-4c7a-9efa-ea90c8c59c03?lng=en>



Full Stack Dart

with Tony Pujals and Kevin Moore

<https://gcppodcast.com/post/episode-261-full-stack-dart-with-tony-pujals-and-kevin-moore/>

<https://pinboard.in/u:cpswan/t:dart>

★ [yaml_edit | Dart Package](#) ✓

[dart](#) [yaml](#)

29 days ago by [cpswan](#) [edit](#) [delete](#)

★ [WTF are Dart and Flutter?](#) ✓

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★ [Microbenchmarking Dart \(Part 1\)](#) ✓

[dart](#) [benchmark](#)

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★ [Experimenting with compiling Dart to Wasm, and calling Wasm from Dart](#) ✓

[dart](#) [wasm](#)

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★ [Concurrency in Server-Side Dart](#) ✓

[dart](#)

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★ [Iakos - Visualize Dart library dependencies in Graphviz](#) ✓

[dart](#)

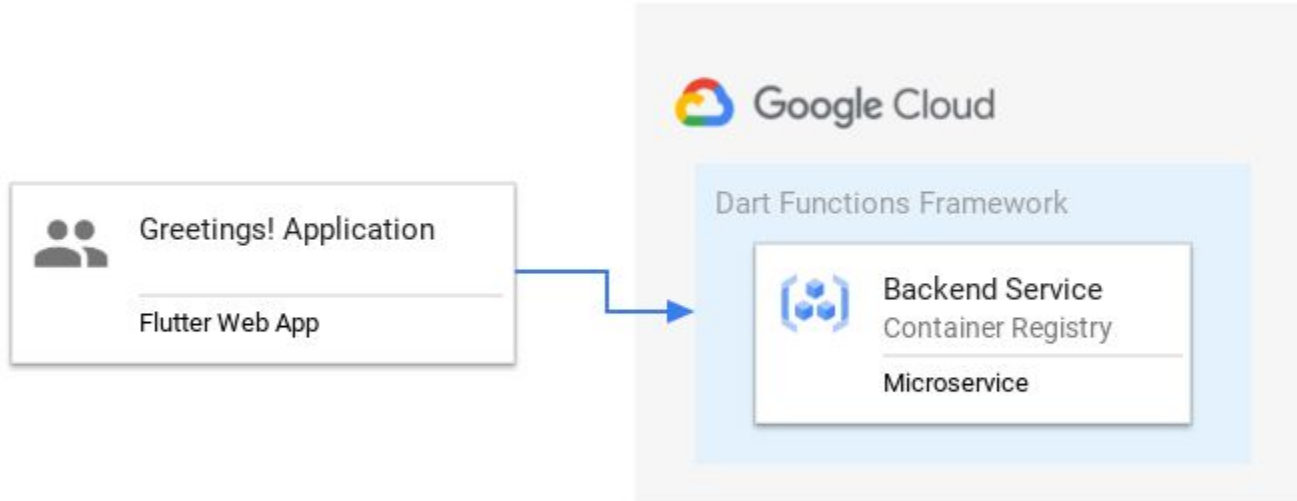
july 2021 by [cpswan](#) [edit](#) [delete](#)



Call to action:
Try the Functions Framework Examples

Qwiklabs - Dart Functions Framework

Dart Functions Framework



Free Qwiklabs from Cloud Skills Challenge

Start your skills challenge today

Choose a challenge track and race the clock to earn your recommended skill badges. Be sure to complete the challenge in 30 days with your free access. Unlock a new challenge every month.

Start challenge



<https://inthecloud.withgoogle.com/google-cloud-skills/register.html>



Review

- Why Dart?
- Language features
- JIT vs AOT
- Dart on Docker
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Questions?

Contact

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@cpswan



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The @ Company

At The @ Company (The At Company) we are technologists, creators, and builders with one thing in common: We love the Internet. You could go so far as to call us Internet optimists. Though we acknowledge that the Internet has deep flaws, we believe that we can extract all its goodness without sacrificing our privacy, time, and control over our digital identities.

We've committed ourselves to the creation of a more human Internet where privacy is a fundamental right and everyone owns their own data. Let's say goodbye to the fear and paranoia caused by data breaches and unsolicited online surveillance.

We've developed a fully-secure, privacy-compliant technology powering customer mobile applications: the @protocol, and with unique identifiers called @signs that act as keys into these experiences. With like-minded partners we're resolving these long-standing issues with a spirit of fun, not fear, and with delightful customer experiences.

FACT SHEET

Founded : 2019, by Barbara Tallent, Colin Constable, and Kevin Nickels, 30+ years executive experience at NCD/Netmanage, Credit Suisse, Deutsche Bank, Juniper Networks, Founded 3 prior startups, 3 exits. **Chairman**: Kim Perdikou, PCNET, Reader's Digest, most recently CIO, GM, EVP Juniper Networks

HQ : Virtual offices, with base in San Jose, CA.

Funding : \$11M seed, May, 2021

Employees: 22

Patents : 15 pending

Website : <https://atsign.com>